

Task 1: Analyse the Scenario and Design the System

a. Incidence-Response responsibilities

CampusGuard supports an incident from registration until it is closed. An incident is registered against a campus area in the **New** state. The operator can then:

- issue a public alert;
- dispatch a response unit to that area;
- change access to the area (lock, unlock, or restrict).

Dispatching a unit coordinates the other response services (security, medical, facilities, communications) so those units do not call each other. Incident status and area access change through defined states; invalid changes are refused.

A high-level workflow can run several of these steps in sequence (activate or resolve). Alerts leave the system through the existing legacy alert interface. When the incident is **Resolved**, the area can be unlocked and an all-clear issued.

b. UML Diagram

Uml segment on showing Command, Mediator and Adapter pattern in our system

Uml segment on showing the different areas, access state and the incident class with it's related states

Uml segment showing Facade pattern in our system

c. GoF Participants in named patterns

I. Command

- Invoker: **Operator**
- Receiver: **CampusArea, TeamCoordinator, AlertService**
- Command: **Command**
- ConcreteCommand: **LockArea, UnlockArea, RestrictArea, DispathUnit, IssueAlert**

II. Mediator

- Mediator: **Coordinator**
- ConcreteMediator: **TeamCoordinator**
- Colleague: **ResponseUnit**
- ConcreteColleague: **CommunicationsTeam, SecurityTeam, FacilitiesTeam, MedicalTeam**

III. Facade

- Facade: **CampusQuardFacade**

IV. Adapter

- Adapter: **AlertServiceAdapter**
- Adaptee: **LegacyAlertService**
- Target: **AlertService**

V. State

- State: **IncidentState, AccessState** ConcreteState: **LockedState, OpenState, RestrictedState, NewState, AssignedState, MitigationState, ResolvedState**
- Context: **CampusArea, Incident**

VI. Composite

- Component: **CampusArea**
- Composite: **Building**
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Leaf: Room

Task 2: Implement the Integrated Model